



HTML5 Local Storage Offline Web Applications

This article explains the local (client-side, browser) storage feature in HTML 5.

Since the introduction of HTML5, the technology is making an equal impact in the desktop and mobile space. It has completely changed what is possible with a Web interface. The new features are so powerful they bring Web pages closer to Apps. The new features include a plug-in-free paradigm, with the introduction of tags like `<video>`, semantic mark-up like `<header>`, new form elements like *email*, and client-side storage. There is a constant drive to make browser-specific incompatibilities disappear.

The need for client-side storage has been felt with the increasing use of Web for e-commerce, with the aim to enhance the user experience by storing user preferences, shopping cart items and user-session information. Cookies have served Web developers for long, but have size limitations (4 KB each) and need to be transferred from client to server. The importance of offline storage has increased with Web app development using HTML5. There are times when the user is not connected to the Internet (e.g., composing an email in an airplane using Internet web app to send later), or when data needs to be saved across sessions (e.g., a Web gaming app needs to store the session state, like the board positions for chess, so that the game can be resumed later).

With HTML5, two new objects are introduced for client-side storage. They are `localStorage` and `sessionStorage`. We will focus on `localStorage` in this article. New object `sessionStorage` is to store data for one session. Before introduction of `localStorage` in HTML5, developers had to choose from various vendor- or browser-specific options like Google Gears, `userData`

in IE, and Local Shared Objects used in Adobe Flash Player.

An introduction to `localStorage`

`localStorage` is also called Web Storage or DOM storage. It is a database that can store only key-value pairs, and the value data-type should always be a string. At first, this might seem a serious limitation, but using along with JavaScript arrays and JSON we can achieve quite a bit. The good news is that this feature is supported in the latest versions of all browsers, both on desktop and mobile environments (Android and iOS). Also, the developer gets a huge 5 MB of storage per domain.

The code snippet below checks if the browser supports `localStorage`. The code in this article has been tested on Google Chrome 17.0.963.56 and Firefox 11.0. There is a small difference in behaviour when the code below is executed in IE: if the HTML file is opened locally (`file:///C:/Web/mylocaltodo.html`), IE throws an error—but if the same file is hosted on a server (even `localhost`), it just works!

In the following code, `localStorage` is an object under `window` and can be referenced via the `window` object, or directly as `localStorage`.

```
function browserCheck() {
    if (typeof(localStorage) == 'undefined') {
        alert('Your browser does not support HTML5 localStorage. Try upgrading.');
```